

Machine Learning Assignment 1

Polynomial Regression

Roll Number: BT2024129

GitHub Repository:

<https://github.com/muzhaib/BT2024129-ML-Assignment-1>

1. Introduction

This assignment investigates polynomial regression for two personalized regression problems, VAR 1 and VAR 2. The objective is to construct polynomial regression models, select suitable polynomial complexity using cross-validation, and generate predictions for the hidden test datasets.

For VAR 1, the training data contains six input features $x_1, \dots, x_6$ and a continuous target y . Polynomial degrees up to 10 were investigated. For VAR 2, the training data contains three input features x_1, x_2, x_3 and the same target structure, with polynomial degrees up to 20 considered.

The experiments used polynomial feature expansion followed by feature scaling and regularized linear regression models. Ridge Regression and Lasso Regression were investigated, with additional Elastic Net experimentation for VAR 2.

2. Approach

2.1 Data Preparation

The training datasets were loaded using pandas. The column y was used as the target, while the remaining columns were treated as input features.

The datasets contained 1000 training samples. The corresponding hidden test datasets contained 1000 samples each. Before model fitting, the input variables were transformed using polynomial feature expansion. The experiments used:

- PolynomialFeatures
- StandardScaler
- Ridge Regression
- Lasso Regression
- Elastic Net for an additional VAR 2 experiment

The complete preprocessing and regression operations were implemented using scikit-learn pipelines.

2.2 Polynomial Feature Expansion

For a polynomial model of degree d , the original features are expanded to include polynomial and interaction terms whose total degree is at most d .

For example, with two variables, a degree-three expansion contains terms such as

$$1, \quad x_1, \quad x_2, \quad x_1^2, \quad x_1x_2, \quad x_2^2, \quad x_1^3, \quad x_1^2x_2, \quad x_1x_2^2, \quad x_2^3.$$

The number of generated features increases rapidly as the polynomial degree and number of input variables increase. Therefore, model complexity and computational cost were both considered when selecting the final models.

2.3 Regularization

Ridge Regression was used to reduce the effect of large model coefficients. Its objective can be expressed as

$$\min_{\beta} \left[\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_j \beta_j^2 \right].$$

Lasso Regression uses an L_1 penalty:

$$\min_{\beta} \left[\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_j |\beta_j| \right].$$

Regularization is particularly useful for polynomial regression because polynomial expansion can create many correlated features and can lead to overfitting.

3. Model Selection and Validation

Five-fold shuffled cross-validation was used for model selection. The experiments used

```
KFold(n_splits=5, shuffle=True, random_state=42)
```

together with `GridSearchCV`.

The primary model-selection metric was mean squared error (MSE). Since scikit-learn’s grid search maximizes a scoring value, negative MSE was used internally.

The cross-validation procedure was preferred over evaluating only the training error because polynomial models can fit the training data very closely while performing poorly on unseen data.

The notebooks do not calculate cross-validation R^2 . Therefore, no cross-validation R^2 value is reported in this document. Where training R^2 is shown, it is explicitly identified as training performance rather than validation performance.

4. Computational Considerations

Polynomial regression becomes computationally expensive as the number of input features and polynomial degree increase.

VAR 1 has six input features. Consequently, high-degree polynomial expansion produces substantially more terms than the corresponding expansion for VAR 2, which has only three input features.

The broader polynomial and regularization searches required significant computation, with some experiments taking more than two hours. To keep the experiments computationally manageable, the final notebook experiments used the following practical search ranges:

- VAR 1 Ridge: polynomial degrees 1–10.
- VAR 1 Lasso: polynomial degrees 1–6.
- VAR 2 Ridge: polynomial degrees 1–20.
- VAR 2 Lasso: polynomial degrees 4–13.
- VAR 2 Elastic Net: polynomial degrees 10–14.

The use of regularization also helped control the increased complexity introduced by polynomial expansion.

5. VAR 1

VAR 1 contains six input features and allows polynomial degrees up to 10. The experiments first investigated Ridge Regression over different degrees and regularization strengths. The regularization parameter α was subsequently refined around promising values.

5.1 Ridge Regression Search

The main Ridge experiments produced the following results.

Experiment	Degree	α	CV MSE
Initial Ridge search	5	10.0	0.534843
Fine α search 1	5	15.0	0.520518
Fine α search 2	5	21.0	0.516776
Final degree search	5	15.5	0.519840

The Ridge experiments consistently selected degree 5 as the best polynomial degree. However, Lasso Regression achieved a substantially lower cross-validation MSE.

5.2 Lasso Regression

The Lasso search investigated polynomial degrees and different regularization strengths. The best configuration used:

$$\text{Degree} = 5, \quad \alpha = 0.01389495494$$

with a cross-validation MSE of

$$\text{CV MSE} = 0.3381889241.$$

The corresponding training R^2 was approximately

$$R_{\text{train}}^2 = 0.97295.$$

The Lasso model was therefore selected for final VAR 1 inference.

The choice of degree 5 was supported by the cross-validation experiments, while the Lasso penalty provided better validation performance than the Ridge configurations tested.

6. VAR 2

VAR 2 contains three input features and permits polynomial degrees up to 20. Since the number of original features is smaller than in VAR 1, higher polynomial degrees can be explored with a lower feature-expansion cost.

6.1 Ridge Regression Search

The Ridge experiments produced the following results.

Experiment	Degree	α	CV MSE
Initial Ridge search	12	1.000	0.259246
Fine α search 1	12	1.815	0.256677
Fine α search 2	12	1.865	0.256674
Final degree search	12	1.865	0.256674

The Ridge experiments consistently identified degree 12 as the best degree in the tested range.

6.2 Lasso Regression

Lasso Regression was then evaluated over polynomial degrees and regularization strengths.

The best configuration was

$$\text{Degree} = 10, \quad \alpha = 0.000316227766$$

with

$$\text{CV MSE} = 0.2531166843.$$

Its training R^2 was

$$R_{\text{train}}^2 = 0.996510.$$

The Lasso model was therefore selected for the final VAR 2 test-set predictions.

6.3 Elastic Net Experiment

An additional Elastic Net experiment was performed after the Lasso-based test predictions had already been generated. The search considered polynomial degrees 10–14, several values of α , and l_1 ratios of 0.2, 0.5 and 0.8.

The best configuration from this additional experiment was

$$\text{Degree} = 14, \quad \alpha = 0.01, \quad l_1\text{-ratio} = 0.2.$$

Its reported cross-validation MSE was approximately

0.285891.

This result was worse than the selected Lasso model's CV MSE of approximately 0.253117. More importantly, the Elastic Net experiment was performed after the Lasso test predictions had been generated. Therefore, Elastic Net was not used for the final VAR 2 submission.

7. Final Results and Discussion

The final configurations used for inference are summarized below.

Problem	Model	Degree	α	CV MSE
VAR 1	Lasso	5	0.013895	0.338189
VAR 2	Lasso	10	0.000316	0.253117

For both problems, Lasso Regression produced the best validation performance among the final models used for inference.

The difference in selected polynomial degree is reasonable given the different feature dimensions and the observed validation performance. VAR 1 contains six input variables, so increasing the polynomial degree causes a much larger expansion of the feature space. VAR 2 contains three input variables, allowing higher-degree models to be explored more easily.

The selected models were fitted using the training data and then used to generate predictions for the corresponding hidden test datasets. Since the true labels of the hidden test data are unavailable, a test-set MSE or R^2 cannot be calculated locally.

8. Inference and Prediction Files

After selecting the final models, predictions were generated for the provided test datasets.

For VAR 1, the final Lasso model with degree 5 was used.

For VAR 2, the final Lasso model with degree 10 was used. The later Elastic Net experiment was not used for the submitted predictions.

Each prediction file contains one column, y , corresponding to the predicted target values.

The expected submission filenames are:

- BT2024129_pred_var1.csv
- BT2024129_pred_var2.csv

Both prediction datasets contain 1000 predicted values.

9. GitHub Repository

The complete implementation, including the polynomial regression notebooks, prediction files, and report, is available in the following GitHub repository:

<https://github.com/muzhaib/BT2024129-ML-Assignment-1>

The repository contains the code used for the experiments and inference along with the generated assignment outputs.

10. Conclusion

Polynomial regression was investigated for both personalized regression problems using cross-validation and regularization.

For VAR 1, the final selected model was Lasso Regression with polynomial degree 5 and $\alpha \approx 0.013895$, achieving a cross-validation MSE of approximately 0.338189.

For VAR 2, the final selected model was Lasso Regression with polynomial degree 10 and $\alpha \approx 0.000316$, achieving a cross-validation MSE of approximately 0.253117.

The experiments demonstrate the importance of balancing polynomial model complexity with regularization. Higher polynomial degrees provide additional flexibility but also increase the number of generated features and computational requirements. Cross-validation and regularization were therefore used to select models that generalized better than simply minimizing training error.